SAINT MICHAEL COLLEGE OF CARAGA

CHALLENGES ENCOUNTERED BY FIRST RESPONDERS IN DOCUMENTING CRIME SCENE

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

The documentation of crime scenes is an essential component of criminal investigations and serves as the foundation for successful prosecutions. Effective documentation allows investigators to “freeze time” and capture details that might otherwise be lost, transforming physical evidence into a reliable narrative (N. Kumar et al., 2023; Singh, 2021). Crime scene officers must possess the necessary knowledge and skill to preserve the integrity of evidence (Upasana Borah et al., 2020). Maintaining record integrity ensures the quality of evidence and supports accurate analysis, leading to reliable results and just outcomes (Heavey et al., 2023; Lawan & Henttonen, 2024). However, poor documentation remains a global concern, with systemic issues such as inadequate recording, improper handling, and organizational deficiencies contributing to forensic errors in nearly half of wrongful conviction cases (Morgan et al., 2023). These challenges highlight the need for standardized procedures, comprehensive training, and improved documentation protocols to maintain investigative accuracy and public trust (Heavey et al., 2023).

Globally, first responders face numerous challenges in maintaining accurate and reliable crime scene documentation (Kösem et al., 2022). Incomplete or erroneous documentation can lower prosecution success and undermine credibility (Magahed, 2021). Effective records should include notes, photographs, sketches, videos, measurements, and formal reports (Sharma et al., 2019). Yet many countries struggle

with limited funding, outdated equipment, and insufficient forensic tools, which hinder evidence preservation and increase the risk of contamination or loss. Environmental factors further complicate documentation: heat, humidity, heavy rain, and strong winds can degrade biological evidence, accelerate decomposition, and damage crime scenes (Sacco et al., 2024; Khorwal et al., 2024; Ashbridge et al., 2022). High temperatures alone may increase tissue decay by up to 33.5%, and decomposition, insect activity, and other environmental exposures can obscure critical forensic details (Strack et al., 2023; Ziogos et al., 2023).

In the Philippines, law enforcement faces similar challenges. Limited resources, outdated tools, minimal training, and environmental factors such as poor lighting and weather hinder evidence preservation (Riambon et al., 2025; Sai Yrkm et al., 2022). Coordination issues and delayed responses can further lead to evidence contamination or loss, while insufficient training and inadequate storage undermine investigative credibility (Thompson et al., 2022; McCartney et al., 2023). These challenges weaken public confidence in the legal system. Given the country's exposure to natural disasters and growing crime complexity, effective and adaptive documentation techniques are essential for ensuring accurate investigations and justice delivery (K. Iuchi et al., 2019; Quiape et al., 2023).

This study aims to address the challenges faced by first responders in documenting crime scenes effectively. It examines constraints such as limited resources, lack of training, environmental hazards, and coordination problems that hinder accurate evidence preservation. It also seeks to identify best practices and strategies to improve

the efficiency and accuracy of crime scene documentation in the local context. By focusing on these areas, the study contributes to enhancing investigative processes and ensuring reliable criminal justice outcomes.

The significance of this research lies in its potential to strengthen criminal investigation practices. Findings will guide law enforcement agencies, policymakers, and educational institutions in designing training programs, allocating resources, and formulating strategies that enhance first responders' capabilities. Improved skills and preparedness in documenting crime scenes will lead to more accurate investigations, fairer prosecutions, and greater public trust in the justice system.

1.2 Review of Related Literature

Age

Age plays a crucial role in shaping the performance of police officers in documenting crime scenes. Khalilia et al. (2024) found a statistically significant difference ($p < 0.05$) in crime scene processing experiences related to officer age among 244 Palestinian police crime scene investigators and officers. Their study examined documentation, collection, preservation, packaging, and transportation of biological traces, suggesting that age can influence performance in these critical tasks.

Beyond procedural skills, the maturity and experience associated with age can affect officers' approach to crime scene management. Miano et al. (2024) explained that older officers often exhibit greater professional maturity, procedural consistency, and measured decision-making, which contribute to more effective and accurate documentation practices.

Moreover, physical fitness, which tends to decline with age, also plays a significant role in crime scene documentation. Frick et al. (2024) reported that police officers experience more pronounced decreases in cardiovascular health, muscular strength, and mobility compared to the general population as they age. These physiological changes can affect an officer's ability to maneuver around complex or chaotic crime scenes, potentially influencing the accuracy and completeness of evidence collection and documentation.

Ultimately, age impacts both the cognitive and physical capacities of first responders, shaping overall efficiency, accuracy, and decision-making during crime scene documentation. Recognizing the interplay between experience, professionalism, and age-related physical changes is essential for designing targeted training programs and support strategies. By considering age as a factor, law enforcement agencies can ensure that officers across all age groups maintain high standards in evidence documentation and investigative performance (Khalilia et al., 2024; Miano et al., 2024; Frick et al., 2024).

Sex

Sex plays a critical role in shaping how first responders perform and experience crime scene documentation. Gregory et al. (2025) emphasized that emergency response work remains male-dominated, with workplace structures and cultures often designed around men. Although women are increasingly entering professions such as policing, firefighting, and paramedicine, they continue to face gendered challenges, including sexism, unequal promotion opportunities, and limited access to proper equipment. Despite these barriers, female first responders frequently demonstrate strengths such as

emotional intelligence, effective communication, and conflict de-escalation skills, which can enhance investigative and documentation accuracy.

Supporting these findings, Degerstedt et al. (2025) observed that female police officers encounter unique physical and cultural challenges, including ill-fitting uniforms, limited promotion opportunities, and difficulties balancing work with family responsibilities. Similarly, Marshall et al. (2021) reported nuanced gender differences among nine Australian crime scene examiners (five male, four female), noting that female examiners faced challenges related to rank and organizational treatment, and those with caring responsibilities experienced negative career impacts when working part-time. These studies suggest that organizational culture and gender dynamics can indirectly influence how documentation duties are performed and perceived.

Related research presents mixed findings regarding sex and overall law enforcement performance. Lockie et al. (2020) found that female recruits performed slower on physical job tasks, while Desmond et al. (2020) observed that female officers were rated lower in overall performance evaluations, though documentation-specific tasks were not examined. Gendered expectations and societal perceptions can further shape officer behavior and decision-making. Vieira et al. (2022) demonstrated that male and female officers held differing beliefs about gender roles in intimate partner violence interventions, affecting their actions and empathy toward victims. Similarly, Gaub (2019) and Navarro et al. (2025) noted that officer sex can influence public perceptions and contextual expectations, such as a higher likelihood of recommending backup for female officers, though broader performance outcomes were not consistently affected.

Overall, these studies highlight the complexity of understanding sex-related differences in policing and forensic work. While female officers demonstrate equal professional capability, institutional biases, cultural expectations, and gendered workplace structures continue to shape their experiences, opportunities, and potentially the quality and reliability of crime scene documentation. Comprehensive, data-driven research is necessary to determine how sex-based differences, if any, directly influence documentation outcomes.

Rank

Ranks play an important role in how first responders do their work during crime scene documentation. According to Davis (2018), an officer's rank shows how much power, responsibility, and decision-making they have at a crime scene. Higher-ranking officers usually lead the process and make sure that all steps are done correctly. Meanwhile, lower-ranking officers handle the tasks like collecting evidence, taking photos, and writing reports. This kind of structure helps keep the work organized, but it can also cause delays if people from different ranks don't communicate well.

Selby et al. (2022) explained that an officer's rank also affects their access to training and resources. Senior officers often get more advanced training and better equipment, helping them keep the documentation accurate. On the other hand, lower-ranking officers may have limited tools and less training, which can lead to mistakes in recording or keeping evidence safe.

Brian and Normore (2019) added that rank influences how well first responders work together. Good leadership from higher-ranking officers can improve teamwork,

reduce repeated tasks, and avoid mistakes. But if leaders fail to communicate clearly, it can harm the accuracy of documentation and increase the risk of losing or mishandling evidence.

Marshal et al. (2021) also found that rank and police culture affect how confident lower-ranking officers feel. In strict organizations, junior officers may wait for orders before taking action, which can slow down the process. But in teams where they are encouraged to take initiative, documentation tends to be faster, more accurate, and more consistent.

In short, ranks affect how much control, responsibility, and skill each first responder has in documenting a crime scene. Good communication, fair training, and strong supervision are all important to make sure the evidence is recorded correctly and reliably.

Length of service

According to Somers et al. (2022), length of service is a key demographic factor influencing police officer performance, confidence, and operational efficiency. They found that officers with longer tenure reported significantly higher confidence in performing job-related tasks, suggesting that experience contributes to self-assurance in complex situations. However, research indicates that tenure alone is not a straightforward predictor of effectiveness, as performance outcomes can vary depending on the context of work and individual coping strategies.

Building on this, Baldwin et al. (2022) examined 122 officers who operated in high-stress scenarios and observed that the length of police service had complex effects on both performance and errors involving lethal force. Similarly, Fedorenko et al. (2020) found that novice officers (with three months of experience) and experienced officers (with 5–15 years) employed similar coping strategies, highlighting that experience does not automatically translate into superior performance in every aspect of policing. These studies suggest that organizational support, training, and situational context are as important as experience for operational success.

Complementing these insights, Craven et al. (2022) identified resilience, emotional regulation, self-efficacy, and optimism as critical factors for effective performance among crime scene investigation personnel. Even experienced officers benefit from developing personal and professional coping skills, which help them manage stress, maintain focus, and navigate complex crime scene situations effectively.

The Abu Dhabi Crime Scene Department demonstrates how length of service affects police work. Most officers there have 11 to 15 years of experience, while smaller groups have less than 5 years or between 6 and 10 years of service (Rashed Saleh Al Shamsi et al., 2023). This indicates that mid-career officers form the operational backbone of the department, combining sufficient experience to handle complex tasks with adaptability to evolving practices and technology.

Building on this foundation, Hawthorne (2021) underscores that longer length of service allows officers to apply learned skills consistently, improving confidence and procedural efficiency at the crime scene. The integration of formal education with on-

the-job experience enables officers to navigate complex forensic scenarios, make informed decisions, and maintain evidence integrity. This demonstrates that experience gained over years of service complements formal education, enhancing overall operational effectiveness.

Mofokeng and Aphane (2022) also explored knowledge management capabilities among South African Police Service officers during crime scene processing. Their study included participants with varying levels of experience, from new recruits to highly competent officers, and demonstrated that length of service influenced awareness of the crime scene, identification of appropriate expertise, and effective control of the scene. Officers' tenure affected their ability to navigate procedural challenges, coordinate with colleagues, and reduce the risk of evidence being rendered inadmissible, emphasizing that length of service contributes to practical decision-making and procedural efficiency in forensic contexts.

Furthermore, Kelty et al. (2024) highlighted that top-performing crime scene examiners exhibit a combination of professional demeanor, cognitive abilities, and life/work experience, suggesting that length of service paired with holistic skill development enhances operational efficiency and reduces errors at crime scenes. Similarly, Knes et al. (2023) emphasized that crime scene investigators' decision-making varies across stages of an investigation and that experience interacts with prior knowledge, contextual understanding, and hypothetico-deductive reasoning to influence outcomes.

Finally, Mejia (2025) found that length of service influenced how officers perceived challenges in technological use, coordination, time management, and logistical factors during crime scene operations. Officers with longer service were better equipped to manage operational efficiency, evidence analysis, coordination, and court testimony, suggesting that tenure contributes to both confidence and procedural competence. However, minor challenges persisted across all experience levels, indicating that operational effectiveness depends on more than just length of service. Likewise, Lidén and Almazrouei (2023) demonstrated that even experienced crime scene investigators' decisions regarding selection and prioritization of blood traces can vary depending on situational factors such as awareness of limited resources and contextual information indicating homicide or suicide.

Together, these studies highlight that while length of service provides valuable procedural knowledge and confidence, effective decision-making at the scene is also shaped by external conditions. This emphasizes the importance of continuous training, standardized protocols, and operational guidance to support consistent and reliable evidence collection.

Level of training related to crime scene documentation

The Philippine National Police (PNP) believes that proper training is the foundation of good crime scene documentation and investigation. Based on the *PNP Manual Revised Directorate for Investigation and Detective Management Administrative and Operations Manual (2020)*, the PNP continues to improve the skills of its officers through special training programs like the Crime Scene First Responders Workshop

(CSFRW) and the Criminal Investigation Course (CIC). These programs help officers learn how to handle, record, and protect evidence properly. Another program, the Basic Scene of Crime Operations Course (BSOCOC), was introduced in 2024 and is managed by the *National Forensic Science Training Institute (NFSTI)* under the *Philippine Public Safety College (PPSC)*. This course focuses on real field practice and forensic training. These programs show that the PNP values continuous learning and skill development so that officers are always ready, knowledgeable, and capable of documenting crime scenes accurately. This proves that training and education are the strong foundation of effective investigation and evidence protection in the Philippines.

Training plays a very important role in how first responders record, protect, and understand evidence at a crime scene. According to Kumar and Singh (2023), good crime scene documentation needs both knowledge and practical skills, which can only be developed through continuous and specialized training. Officers who are properly trained in taking photos, making sketches, measuring, and collecting evidence are more accurate and consistent in their work. On the other hand, limited or outdated training can lead to mistakes, missing records, and damaged evidence.

Daibes et al. (2025) also explained that structured forensic education programs help officers follow standard rules and make better decisions in stressful situations. Their study showed that officers with forensic training were more likely to maintain the chain of custody and follow standard operating procedures (SOPs).

Meshal Albeedan et al. (2024) found that many problems in crime scene documentation come from a lack of real-life practice. When training focuses only on

theory and not on hands-on experience, responders may struggle in difficult situations like bad weather, poor lighting, or crowded areas. Training that includes real field practice and the use of modern tools like digital photography, 3D scanning, and body cameras greatly improves the quality of documentation and awareness of the situation.

Noh and Kim (2024) added that continuous professional training is very important because crime scene procedures and technology are always changing. Regular refresher courses help responders stay updated and prevent mistakes that could contaminate or misinterpret evidence. Similarly, Nilendu (2024) pointed out that effective training should not only teach technical skills but also include communication, teamwork, and ethical awareness, since these all affect how accurate and reliable the documentation will be.

In summary, the level of training directly affects how skilled, confident, and efficient first responders are in documenting crime scenes. Continuous and well-designed training helps them handle complex situations, use new technology properly, and protect the integrity of evidence. On the other hand, poor or outdated training can lead to errors, poor coordination, and incomplete documentation issues that this study aims to explore.

Adequacy of tools and equipment

Having the right tools and equipment is very important for first responders to document crime scenes accurately and efficiently. According to Megahed (2022), proper documentation does not only depend on the investigator's skill but also on the availability of useful tools like cameras, sketching kits, measuring devices, gloves, and

evidence bags. When officers do not have the right equipment or use old or broken tools the chance of making mistakes, losing evidence, or recording incomplete information becomes much higher.

Galvin (2020) also explained that the success of crime scene documentation depends on using the correct forensic instruments to take clear photos, accurate measurements, and organized notes. If resources are lacking, it becomes harder to record observations properly, which weakens the value of the evidence presented in court. Even well-trained officers can struggle to stay accurate and maintain proper handling of evidence when their tools are limited.

Mateen and Tariq (2019) found that in many developing countries, first responders often face problems because of a lack of equipment and technical resources. These shortages cause delays and force officers to rely on manual methods, which lowers the quality of written and visual evidence. This makes it harder to rebuild or understand what really happened at the crime scene during investigations or trials.

Smith and Horne (2024) reported that many police agencies have small budgets, leading to a lack of important forensic tools. This shortage not only affects the quality of documentation but also lowers the motivation and performance of officers. Their study suggested that agencies should regularly check their equipment and make sure all investigators have the tools they need.

Deshmukh et al. (2023) mentioned that modern technology like 3D scanning and virtual reality tools have greatly improved crime scene documentation by creating more accurate digital records. However, many local police stations cannot afford these new

tools or lack the training to use them. Adopting modern technology helps make data more precise, supports evidence reconstruction, and reduces human mistakes.

In conclusion, having enough and properly working tools and equipment is essential for accurate crime scene documentation. Without updated and well-maintained resources, even skilled officers may struggle to collect and record evidence correctly. Providing complete and modern equipment, along with regular maintenance and training, is necessary to improve the quality and reliability of crime scene investigations.

Knowledge and technical skills

The knowledge and technical skills of first responders are very important in making sure that crime scenes are documented correctly and clearly.

According to Daibes et al. (2025), documenting a crime scene needs both classroom knowledge and practical skills like taking clear photos, making accurate sketches, measuring properly, and keeping evidence safe. Responders should know how to observe, record, and explain what happened at the scene. If they lack these skills, the evidence might be wrong, incomplete, or not accepted in court.

Illes et al. (2021) explained that doing good crime scene work is not just about experience. It also requires proper education, training, and clear thinking. Responders need to understand the science behind documentation to avoid mistakes. Those who depend only on experience might miss small but important clues or use the wrong methods.

Brown et al. (2023) said that combining knowledge with critical thinking is key to good documentation. Investigators who have both understanding and practical skills can handle difficult or changing crime scenes better. This helps them make correct decisions and avoid mistakes that could harm the case.

In another study, Kelty et al. (2024) found that responders with strong technical skills were more confident and worked faster when handling complicated scenes. They were also better at dealing with problems like poor lighting or bad weather, which helped them create clearer photos and reports. The study said that continuous training and skill improvement are important to keep investigations accurate.

Rinaldi et al. (2024) also pointed out that knowing how to use modern technology is very important today. Tools like digital cameras, mapping devices, and 3D scanners need special training. Officers who don't know how to use them may make mistakes or leave out details. But those who are skilled can make accurate and detailed digital records that are useful in court.

In short, the knowledge and technical skills of first responders greatly affect how well a crime scene is documented. When responders are well-trained and skilled, they can collect and record evidence correctly. But if they lack knowledge or skill, it can lead to mistakes and damage the investigation. That's why continuous training and education are important to improve crime scene documentation and ensure accurate results.

Coordination and Communication among responders

Good coordination and communication are very important when documenting a crime scene. Police officers, investigators, and forensic teams must work together and share information clearly so that no evidence is lost or damaged. According to Taraskevičiūtė & Dobržinskienė (2023), clear communication helps officers do their job well, avoid confusion, and keep public trust. When responders talk and listen to each other properly, they can solve problems faster and make better decisions.

Brown et al. (2022) said that teams who have open communication and teamwork finish their work faster and make fewer mistakes. If officers don't talk clearly or if there is poor coordination, it can cause delays, double work, or missing evidence. Working together helps everyone understand their role and makes the investigation smoother.

In a study by Tin et al. (2025), weak communication was one of the main problems during emergencies. They found that using digital radios, phones, and clear reporting systems helps responders share updates quickly and work more effectively. Good coordination means everyone knows what to do and when to do it.

Technology can also help improve teamwork. Nelis et al. (2018) showed that tools like 3D scanning and Virtual Reality (VR) let investigators share a realistic view of the crime scene. This makes it easier for different teams to understand what really happened, even if they are not at the scene.

In the Philippine National Police (PNP) Manual (2020), officers are reminded to always communicate with each other, especially between the First Responder, Investigator-on-Case, and the SOCO Team. Using radios, giving updates, and having a clear plan help keep the evidence safe and make the investigation more successful.

In short, coordination and communication help police officers work as one team. When they share information clearly and follow proper procedures, the investigation becomes faster, more accurate, and more reliable. But if they fail to communicate, mistakes can happen, and important evidence might be lost.

Adherence to standard operating procedures (SOPs)

According to Criminal Investigation Manual (CDI-1) on book page 32, Crime Scene Investigation (CSI) is one of the most important parts of a criminal investigation. The first actions of the First Responders, Investigators, and SOCO Team are very important because they can affect the whole outcome of the case. However, this stage is also the most confusing and often full of mistakes because of the pressure and complexity of the scene. Case reviews by the DIDM through the EP JUST Program found that confusion usually happens when the roles of the First Responder (FR), Investigator-on-Case (IOC), and SOCO Team are not clearly defined. To solve this, the PNP Crime Laboratory made clear rules and guidelines to show the duties and responsibilities of each team member. The Investigator-on-Case is mainly in charge of the scene, making sure that evidence is kept safe, witnesses are identified, and all steps are done correctly. Overall, CSI includes all the investigative work from the time the First Responders arrive until the crime scene

is released, showing how important teamwork, organization, and following proper procedures are in solving a case successfully.

The investigation process will begin at the crime scene with the identification and collection of physical evidence. According to Singh (2021), the proper recovery, documentation, and preservation of evidence will be crucial because they will serve as the foundation of the entire investigation. After the evidence and witness statements are analyzed, the findings will later be presented in court as part of the legal process.

Crime Scene Investigation (CSI) is very important to make sure that all evidence is collected and checked properly. Osman et al. (2021) said that the success of an investigation depends on how well officers follow the Standard Operating Procedures (SOPs) these are step-by-step rules that guide them in securing the area, collecting, and recording evidence. But sometimes, officers fail to follow these rules because of lack of training, not enough tools, or poor teamwork. When SOPs are not followed, evidence can be damaged or lost, which can affect the whole case. Following the SOPs helps make sure that the investigation is accurate, trustworthy, and accepted in court.

According to Mousseau et al. (2025), following standard rules in crime scene investigations helps make the work consistent and reliable. However, they also said that being too strict with these rules can sometimes limit the investigator's own judgment. In real crime scenes, every situation is different, so investigators often need to adjust and make smart decisions based on what they see. Because of this, there should be a balance between strictly following the rules and allowing investigators to use their

experience and judgment. This balance helps keep the investigation accurate, trustworthy, and realistic.

Environmental or Situational factors (e.g., weather, crowd control, safety risk);

Environmental or situational factors are conditions in a crime scene that can affect how first responders work, document, and protect evidence. These include situations like a chaotic environment, bad weather, bystander interference, and hard-to-reach crime scenes. Each of these can make documentation harder and can affect how accurate and complete the investigation becomes.

A chaotic crime scene is one of the most difficult situations for first responders to handle because it can cause confusion, contamination, and loss of important evidence. Kösem and Yalçın Sarıbey (2022) explained that when teamwork and proper control are lacking, responders may struggle to organize the area and maintain order. Without coordination, documentation becomes less accurate and the risk of damaging evidence increases.

Another major factor is weather conditions, which can directly affect both the evidence and the performance of the officers. Heat, rain, wind, and humidity can damage or wash away physical evidence. Kowalske et al. (2024) found that high temperatures can change the appearance of bloodstains, while Obradovich et al. (2018) stated that working in extreme heat can make officers tired and less focused. These environmental challenges reduce the accuracy and quality of documentation and may delay the investigation process.

Bystander interference also poses a serious challenge to first responders. Curious or untrained civilians at the scene may distract investigators and accidentally destroy or contaminate evidence. Bruchem et al. (2024) and Andersson and Sundin (2021) pointed out that bystanders who take photos or videos can interfere with official documentation and violate privacy. Their uncontrolled presence can make it difficult for officers to secure the area and properly record what happened.

Lastly, scene accessibility affects how fast and efficiently first responders can reach and document a crime scene. Locations that are far, isolated, or difficult to access can cause delays in evidence collection. Ali Shah and Hussain (2020) observed that the lack of mobile units, proper transportation, and coordination often make it hard for officers to arrive on time, leading to loss or damage of vital evidence.

Another significant improvement is the use of modern technology. Deshmukh et al. (2023) recommended the use of advanced tools such as 3D scanning and Virtual Reality (VR) to make crime scene documentation clearer and more accurate. These technologies allow investigators to recreate the scene digitally, giving them a detailed and distortion-free view that can be analyzed even after the actual site is cleared. Similarly, Krishnan and Zhou (2020) proposed using computer systems that can predict and locate crime scenes faster, helping first responders arrive quickly and work more efficiently during emergencies.

Training and proper equipment are also key to improving performance under difficult environmental conditions. Obradovich et al. (2018) emphasized that officers should be given the right protective gear and regular training so they can perform

effectively even in harsh weather. Well-prepared responders are more capable of maintaining focus, safety, and accuracy while documenting evidence in extreme conditions such as heat, rain, or limited visibility.

Improving community and bystander management can also help minimize interference at crime scenes. Ciottone (2023) suggested that educating and training civilians in basic emergency response can transform them from potential distractions into helpful supporters during critical situations. Through public awareness and cooperation, bystanders can assist responsibly without disrupting official investigations.

Lastly, strong access and evidence preservation protocols are vital for protecting the integrity of a crime scene. Kösem et al. (2022) highlighted the importance of setting clear entry and exit rules to maintain control and reduce contamination risks. Likewise, Sharman et al. (2023) recommended taking quick photographs before any medical treatment begins to preserve key forensic details that could be lost later. These practices ensure that essential evidence remains properly recorded and available for analysis.

In summary, Environmental and situational factors affect how first responders document crime scenes. Problems like a messy or crowded scene, bad weather, bystanders, and hard-to-reach areas can cause confusion, delay, or loss of evidence. These challenges make it harder for officers to stay organized and record information correctly. However, studies show ways to solve these problems. Using modern tools like 3D scanning and Virtual Reality (VR) helps make documentation clearer and more accurate. Giving officers proper training and safety gear also helps them work better in difficult situations. Teaching bystanders how to behave and following clear access rules

keep the scene safe and organized. In short, even if environmental and situational factors cause problems, teamwork, training, technology, and good procedures help first responders do their job well and keep crime scene records accurate and reliable.

Legal or procedural documentation requirements

Legal and procedural documentation rules are the backbone of every crime scene investigation. They set the steps that first responders must follow to make sure that evidence is accurate, protected, and accepted in court.

According to Megahed (2021), proper documentation acts as a legal record that connects the physical evidence to the findings of investigators. If responders fail to follow standard procedures, the investigation can lose credibility, and the evidence might not be allowed in court. That's why understanding and following these legal rules is very important in forensic work.

Verma and Chaudhary (2024) also explained that documentation must follow both science-based and legal standards. This includes taking clear photographs, making detailed sketches, writing notes, and completing chain-of-custody forms. These records help prove that the crime scene was handled correctly. Missing or incorrect details in these documents can cause problems in court and make it harder to prove guilt.

Anna et al. (2023) pointed out that following proper procedures especially the chain of custody is very important for keeping evidence valid. Each piece of evidence must be labeled, sealed, and recorded properly. If there are gaps or missing records, it

can raise doubt in court and affect the outcome of the case. They also said that agencies without clear documentation rules often make mistakes that reduce the accuracy of investigations.

In another study, Daibes et al. (2022) found that many mistakes happen because first responders don't fully understand legal requirements. Some officers may forget to record who handled the evidence or fail to keep an entry log of people at the scene. These errors can damage the trustworthiness of both the investigation and the evidence.

Mateen and Tariq (2019) added that legal documentation can be complicated, especially in places with limited resources. When there's not enough time or administrative support, it becomes hard for responders to write complete reports. They suggested that regular training and supervision are needed to help officers follow proper documentation rules.

In summary, legal and procedural documentation ensures that crime scene reports meet the court's standards for fairness and reliability. Following these rules protects investigators and strengthens the justice process by keeping evidence accurate and traceable. On the other hand, failure to follow them can cause errors, loss of evidence, or even case dismissal. That's why continuous training, awareness, and strict application of documentation standards are essential for successful investigations.

1.3 Theoretical Framework

This study will be anchored on two major theories: the **Information Theory of Criminal Investigation** (Willmer, 1970) and the **Systems Theory** (von Bertalanffy, 1950). These theories will complement each other in explaining both the external and internal challenges that first responders will encounter in documenting crime scenes. The Information Theory will emphasize how effective information flow and evidence management influence the accuracy of documentation, while Systems Theory will highlight how organizational structure, coordination, and available resources affect operational efficiency. Together, these frameworks will provide a comprehensive foundation for understanding the different factors that influence the documentation process.

Information Theory of Criminal Investigation (Willmer, 1970)

The Information Theory of Criminal Investigation views the investigative process as a flow of information between the offender, the crime scene, and the investigator. When a crime occurs, the offender leaves behind physical and trace evidence such as fingerprints, footprints, and other materials that act as signals or messages to be interpreted by investigators. The first responders will serve as the initial receivers and transmitters of this information. Their ability to collect, document, and preserve evidence accurately will determine how efficiently information is communicated throughout the investigation.

However, external factors such as weather conditions, bystander interference, chaotic environments, and limited scene accessibility can create “noise” that disrupts

the information flow. In this study, these environmental and situational factors will be treated as key challenges affecting the accuracy of crime scene documentation. Thus, the Information Theory will help explain how these external disruptions affect the process of collecting and transmitting critical information.

Systems Theory (von Bertalanffy, 1950)

The Systems Theory views an organization as a system made up of interrelated and interdependent components that must work together harmoniously to achieve efficiency. In law enforcement, these components include personnel, equipment, communication systems, procedures, and leadership. Each element contributes to the effectiveness of the overall operation. Failures in coordination, inadequate training, limited tools, or weak adherence to Standard Operating Procedures (SOPs) can disrupt the system and negatively affect documentation accuracy.

Applied to this study, Systems Theory will explain the internal and procedural challenges that first responders will face in documenting crime scenes. These include the adequacy of tools and equipment, knowledge and technical skills, communication among responders, and compliance with SOPs. This theory emphasizes that effective crime scene documentation depends not only on individual competence but also on how well the law enforcement system functions as a coordinated unit.

By combining the Information Theory and Systems Theory, this study will provide a holistic understanding of the challenges encountered by first responders. The Information Theory will address the external and situational challenges that affect information flow, while the Systems Theory will focus on internal and procedural issues

that influence operational performance. Together, these frameworks will serve as the theoretical basis for identifying, analyzing, and proposing strategies to enhance the efficiency and accuracy of crime scene documentation in BuenasCar.

1.4 Schematic Diagram

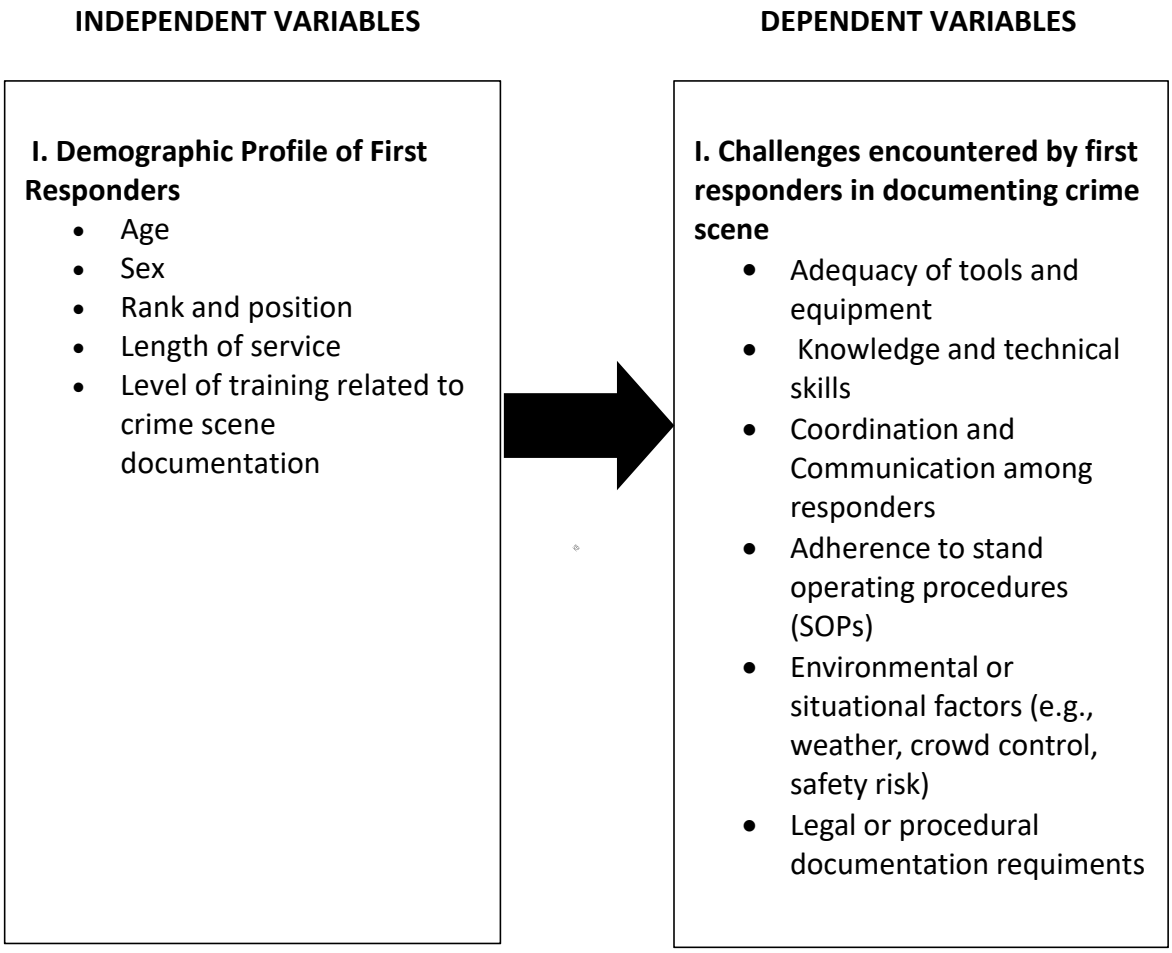


Figure 1. The schematic diagram shows the interplay between independent and dependent variables in the study.

1.5 Statement of the Problem

This study determined the challenges encountered by first responders in documenting crime scenes in BuenasCar. Specifically, it sought to answer the following questions:

1. What is the profile of the respondents in Buenascar in terms of:

   1.1. age;

   1.2. sex;

   1.3. rank; and

   1.4.lenght of service

   1.5. Level of training related to crime scene documentation

2. What is the level of challenges encountered by first responders in documenting crime scenes in terms of:

   2.1. adequacy of tools and equipment;

   2.2. knowledge and technical skills;

   2.3. coordination and communication among responders;

   2.4. adherence to standard operating procedures (SOPs);

   2.5. environmental or situational factors (e.g., weather, crowd control, safety risks); and

   2.6. legal or procedural documentation requirements?

3. Is there a significant relationship between the profile of the first responders and the challenges they encounter in documenting crime scenes?

4. What interventions or strategies may be proposed to enhance the efficiency and accuracy of crime scene documentation among first responders in Buenascar?

1.6 Hypothesis

The hypothesis will guide the study and will be tested at α = 0.05 level of significance:

$Ho1$: There is no significant relationship between the profile of the first responders and the challenges they encounter in documenting crime scenes in Buenascar.

1.7 Definition of Terms

To ensure clarity and consistency throughout the study, the following key terms are operationally defined. These definitions specify how each variable will be understood, measured, and applied within the context of assessing the challenges encountered by first responders in documenting crime scenes in BuenasCar.

Adequacy of Tools and Equipment. Refers to the availability, condition, and functionality of the materials and forensic devices used by first responders in documenting crime scenes. Operationally, it is measured through questionnaire items assessing whether cameras, evidence kits, personal protective equipment, lighting tools, and other documentation materials are sufficient and functional during crime scene operations.

Adherence to Standard Operating Procedures (SOPs). Refers to the extent to which first responders follow the established guidelines, protocols, and documentation standards set by the PNP and Crime Laboratory. Operationally, it is determined through

respondents' rating of their compliance in securing the crime scene, recording evidence, maintaining the chain of custody, and completing required documentation steps.

Age. The chronological age of first responders expressed in years. Operationally, it is grouped by age brackets (e.g., 20–29, 30–39, 40–49, 50 and above) and analyzed to determine its relationship with the challenges encountered in crime scene documentation.

Coordination and Communication Among Responders. Refers to the clarity of roles, teamwork efficiency, and effectiveness of information sharing among first responders. Operationally, it is measured based on the respondents' evaluation of communication flow, task distribution, and collaborative efforts during crime scene documentation.

Environmental or Situational Factors (e.g., Weather, Crowd Control, Safety Risks). The external conditions at the crime scene that may hinder documentation, including weather, lighting, terrain, crowd presence, or safety risks. Operationally, it is assessed through questionnaire items that measure how these factors affect timeliness, accuracy, and completeness of evidence documentation.

Sex. The biological or self-identified classification of respondents as male or female. Operationally, it is used as a demographic variable to determine differences in perceived challenges during crime scene documentation.

Knowledge and Technical Skills. Refers to the first responders' theoretical understanding and practical abilities related to crime scene documentation, such as photographing, sketching, measuring, and evidence handling. Operationally, it is

measured through respondents' self-assessment of their competence and ability to perform documentation tasks correctly.

Legal or Procedural Documentation Requirements. Refers to the mandatory documentation standards required by law and investigative protocols, including reports, photographs, sketches, and chain-of-custody forms. Operationally, it is assessed by how consistently first responders complete these documents according to legal and procedural expectations.

Length of Service. The number of years the respondent has served in law enforcement. Operationally, it is categorized (e.g., 0–5 years, 6–10 years, 11–15 years, 16+ years) and analyzed for its relationship to documentation challenges.

Level of Training Related to Crime Scene Documentation. Refers to the formal education, seminars, workshops, or specialized courses received on crime scene processing. Operationally, it is measured by the respondent's participation in programs such as BSOCOC, CIC, CSFRW, or other PNP forensic training courses.

Rank and Position. Refers to the respondent's official designation within the police force (e.g., Patrolman, Corporal, Sergeant, Lieutenant). Operationally, it is used to determine whether rank influences exposure to or ability to manage challenges in crime scene documentation.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter presents the research methods and procedures used in the study, including the design, materials, participants, data collection, analysis plan, and ethical considerations.

2.1 Research Design

This study will employ a descriptive-correlational quantitative research design**.** Its primary purpose is to identify and analyze the challenges that first responders encounter in documenting crime scenes within the municipalities of Buenavista, Nasipit, and Carmen, Agusan del Norte. The study will focus on various factors influencing documentation accuracy and efficiency, such as adequacy of tools and equipment, adherence to standard operating procedures (SOPs), coordination and communication, environmental and situational factors, knowledge and technical skills, and legal or procedural requirements.

The descriptive aspect will allow systematic documentation of these challenges, while the correlational component will examine statistical relationships between respondents' demographic profiles including age, sex, rank and position, level of training, and length of service and the level of challenges they experience. Through correlation analysis, the study will determine whether specific demographic factors are significantly related to higher or lower levels of difficulty in crime scene documentation.

2.2 Research Materials and Methods

The primary research instrument of this study will be a researcher-made structured questionnaire composed of two parts. The first part will gather the demographic profile of the respondents, including age, gender, rank and position, level of training related to crime scene documentation, and length of service. The second part will measure the challenges encountered by first responders in crime scene documentation in terms of adequacy of tools and equipment, adherence to standard operating procedures (SOPs), coordination and communication among responders, environmental or situational factors, knowledge and technical skills, and legal or procedural documentation requirements.

The instrument will use a 4-point Likert scale: Strongly Agree (4), Agree (3), Disagree (2), and Strongly Disagree (1). Higher scores will indicate greater perceived challenges. The questionnaire will undergo expert validation by professionals in criminology, law enforcement, and research to ensure its validity and reliability. A pilot test will also be conducted prior to full administration. The finalized questionnaires will be personally distributed to respondents after obtaining their informed consent.

2.3 Research Environment, Population and Sample

The study will be conducted in the police stations of Buenavista, Nasipit, and Carmen, Agusan del Norte, which will be selected due to their active engagement in crime scene management and the direct involvement of first responders in documenting incidents. The respondents will consist of police officers who serve as first responders and are directly involved in crime scene documentation.

Table 1. Respondents of the Study

LOCATION	Population	
	F	%
Buenavista	30	37.5
Nasipit	30	37.5
Carmen	20	25.0
Total	**80**	**100.0**

As shown in Table 1, a total of 80 police officers will participate in the study, representing the three municipalities in Agusan del Norte. Thirty officers each will be drawn from the municipalities of Buenavista and Nasipit, accounting for 37.5% each of the total respondents, while twenty officers (25%) will come from Carmen. This distribution will ensure a balanced representation across the three areas, providing a diverse range of perspectives on the operational and procedural challenges that first responders may encounter in documenting crime scenes.

2.4 Research Data and Analysis Plan

The data gathered from the questionnaires will be coded, tabulated, and analyzed using statistical software. Descriptive statistics such as frequency, percentage, and mean will be used to determine the level of challenges encountered by first responders. Correlational analysis, using Pearson or Spearman rho, will be applied to identify the relationship between the respondents' demographic profiles and the challenges they experience in crime scene documentation. The results will reveal the

most common challenges and the possible influence of demographic factors on documentation efficiency.

2.5 Research Ethical Standards

A. Protection of Intellectual Property Rights (IPR)

This research does not involve the development of new technologies, processes, or products that require patent protection, nor does it include any distinctive brands, logos, or symbols warranting trademark registration. However, the study will utilize original materials, such as the structured questionnaire and the final research report, which will be protected under copyright law in the Philippines. The researchers are committed to maintaining the integrity of the study, safeguarding the intellectual property rights of all contributors, and ensuring that the research team and affiliated institutions retain control over the dissemination, publication, and future use of all research outputs.

B. Informed Consent

All written informed consent will be obtained from participants prior to administering the questionnaire. Each participant will be fully briefed on the study's purpose, procedures, and potential risks or benefits. They will have the opportunity to ask questions and clarify any concerns before signing the consent form. Participation will be entirely voluntary, and participants will have the right to withdraw at any time without facing any negative consequences. The researchers will ensure that participants feel informed, supported, and comfortable throughout the data collection process.

C. Data Privacy and Confidentiality

All information collected from the questionnaire will be treated with the utmost confidentiality. Data will be anonymized, coded, and securely stored to prevent unauthorized access. Personal identifiers will be removed to safeguard participants' identities. The researchers will comply with applicable privacy laws and institutional ethical standards to maintain the privacy and confidentiality of all responses.

D. Voluntary Participation and Freedom to Withdraw

Participation in this study will be completely voluntary. Respondents will be clearly informed of their right to withdraw from the study at any stage without repercussions. The researchers will not exert pressure on participants to complete the questionnaire and will encourage them to respond at their own pace, ensuring that participation is fully autonomous.

E. Minimization of Harm and Risk Management

The study focuses solely on collecting data regarding the professional challenges faced by first responders and will not involve any physical, psychological, or emotional harm. Any discomfort or concern arising during questionnaire completion will be addressed immediately. The researchers will ensure participants are fully informed about the study's procedures, risks, and benefits before participation to minimize any potential harm.

F. Beneficence and Contribution to Knowledge

This study adheres to the principle of beneficence by ensuring the welfare of participants while generating valuable knowledge about challenges in crime scene

documentation. The findings are expected to inform law enforcement training, operational procedures, and policy development, promoting more effective and reliable crime scene documentation practices. Results will be shared with participants upon request to ensure transparency and appreciation for their contribution.

G. Justice and Fair Participant Selection

Participants will be selected based on fair and transparent criteria, ensuring that only qualified police officers directly involved in crime scene documentation are included. This process will promote equitable representation and avoid any form of discrimination, aligning with the ethical principle of justice.

H. Data Integrity and Accuracy

The researchers will use a standardized and validated questionnaire to ensure accuracy and reliability of data. All collected data will be carefully reviewed, coded, and analyzed to maintain consistency. Any potential biases, errors, or limitations will be documented and addressed to uphold the integrity and trustworthiness of the study.

I. Transparency and Honesty in Reporting

All findings will be reported truthfully and transparently, regardless of whether they support the initial assumptions. Any limitations or potential sources of error will be acknowledged, and all external sources will be properly cited following academic and ethical standards.

J. Use of Patented or Copyrighted Materials

No patented technologies requiring authorization will be utilized in this research. All copyrighted materials, including academic articles, statistical software, and

references, will be properly cited according to academic standards. Permissions will be obtained when necessary to ensure compliance with copyright and intellectual property laws.

K. Considerations for Human Participants

All participants will be fully informed about the study's purpose, procedures, and potential risks prior to completing the questionnaire. Participation will be voluntary, and respondents may withdraw at any point without consequences. The study will not involve animal experimentation.

L. Responsible Use of AI and Other Technologies

AI tools, including ChatGPT, Perplexity, and QuillBot, will be used responsibly to organize ideas, refine the questionnaire, or verify references. The research team will ensure that all AI-generated outputs are critically reviewed to prevent data misrepresentation or distortion.

M. Ethical Clearance and Institutional Approval

Ethical clearance for the study will be obtained from the Institutional Ethics Review Board of Saint Michael College of Caraga. This approval will ensure that the study complies with institutional, national, and international ethical standards for protecting human subjects, including confidentiality, informed consent, and risk minimization. Documentation of ethical approval will be securely filed for verification and institutional compliance.